

HOIST: Supplementary material

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Abstract

This document provides the HOIST training algorithm, additional placement and payload-motion results, and implementation and hardware details.

HOIST Algorithm

Algorithm S1: HOIST Training

1. **Inputs:** VR teleoperation demonstrations $\mathcal{D}$, pretrained VLA policy π_{VLA} (GR00T N1.6), whole-body controller f_{WBC} (SONIC, frozen throughout)
2. **Output:** Refined hoisting policy $(\pi_{\text{VLA}}^*, \mu_{\theta}^*, f_{\text{WBC}})$
3. **Phase 1 — Supervised Finetuning**
4. Collect VR teleoperation demonstrations with multimodal observations: $\tau_i = (\{o_t, u_t^{\text{demo}}, a_t^{\text{motor}}\}_{t=1}^{T_i}, y_i)$
5. Each observation o_t contains ego RGB/depth, side-view RGB, proprioception, ego velocity, navigation command, and language goal y_i
6. Convert teleoperation commands into VLA action chunks $\Delta U_t^{\text{demo}} = [\Delta u_t, \dots, \Delta u_{t+H-1}]$ with horizon $H = 24$
7. Initialize from the pretrained GR00T N1.6 VLA and attach encoders for newly introduced modalities (depth, ego velocity, navigation command)
8. Supervise the flow-matching action expert to generate demonstration chunks conditioned on VLM features and robot state, using 4 denoise steps
9. Finetune the final 4 VLM layers, the action expert, and the new-modality encoders on $\mathcal{D}$; keep the remaining VLM backbone and f_{WBC} frozen $\Rightarrow$ obtain π_{VLA}^*
10. Freeze π_{VLA}^* and f_{WBC} ; cache VLM features for rollout-level RL refinement
11. **Phase 2 — Iterative Batched RL Refinement**
12. Initialize noise-steering SAC actor-critic (μ_{θ}, Q_{ϕ})

13. For iteration $k = 1, 2, \dots, K$:

1. Collect $N = 10$ autonomous rollouts using $(\pi_{\text{VLA}}^*, \mu_\theta, f_{\text{WBC}})$

2. The actor deterministically predicts a replacement initial-noise tensor

$$a_t^{\text{RL}} = \mu_\theta(o_t), \quad a_t^{\text{RL}} \in \mathbb{R}^{24 \times 25}$$

3. Here, 24 is the action-chunk horizon, and 25 is the per-step planner-command dimension covering head target, left/right end-effector targets, navigation command, and base-height command

4. Replace the flow-matching action expert's initial noise with the scaled actor output:

$$\epsilon_t^{\text{init}} \leftarrow s_k a_t^{\text{RL}}$$

5. Execute the VLA action chunk through the frozen whole-body controller f_{WBC}

6. For each rollout τ_i :

1. Observe terminal payload pose $(\hat{x}, \hat{y}, \hat{\psi})$ and target (x^*, y^*, ψ^*)

2. Compute terminal errors:

$$\Delta x = \hat{x} - x^*, \quad \Delta y = \hat{y} - y^*, \quad \Delta \psi = \hat{\psi} - \psi^*$$

3. Compute terminal reward:

$$r_T = \exp(-[1.5(\frac{\Delta x}{100 \text{ mm}})^2 + 1.0(\frac{\Delta y}{80 \text{ mm}})^2 + 1.0(\frac{|\Delta \psi|}{30})^2])$$

7. Convert rollouts into an offline replay buffer $\mathcal{B}_{0:kN}$

8. Train/update (μ_θ, Q_ϕ) with offline SAC on $\mathcal{B}_{0:kN}$ for fixed gradient steps

9. Resume from previous RL checkpoint for the next stage

14. Return refined hoisting policy $(\pi_{\text{VLA}}^*, \mu_\theta^*, f_{\text{WBC}})$

Design rationale.

During RL refinement, the VLA policy π_{VLA}^* and WBC f_{WBC} remain frozen; only the actor-critic (μ_θ, Q_ϕ) is updated. This preserves the imitation-learned visuomotor prior while restricting exploration to the VLA initial-noise space. Each batch of autonomous rollouts is converted into an offline replay buffer, and RL policy is updated with SAC using terminal task rewards. Thus, the method is not tabula-rasa RL, but rollout-level refinement of the frozen VLA flow-matching action-generation trajectory through a compact noise-steering module.

Additional Placement Results

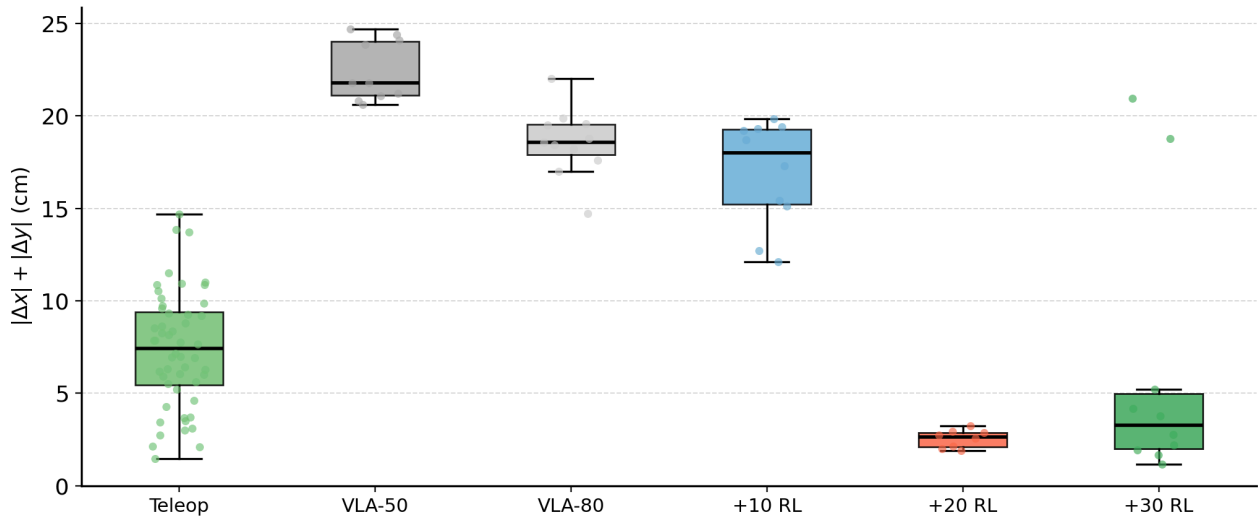
Tables [S1](#) and [S2](#) report terminal placement errors in simulation and on the real platform, respectively. HOIST denotes VLA-50 with iterative batched RL refinement; the rollout count indicates the total RL rollouts used.

Simulation

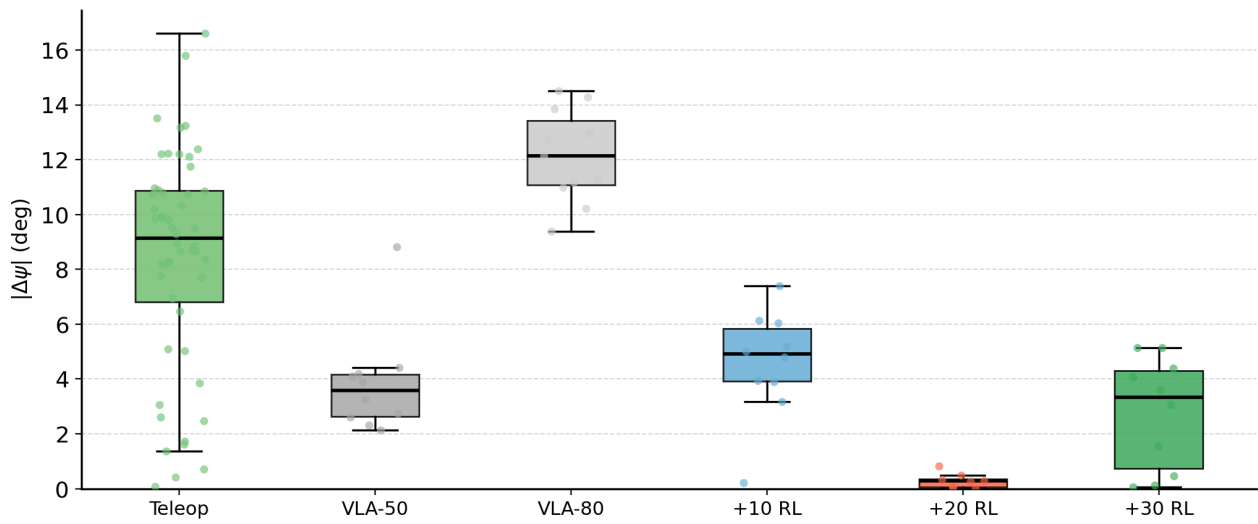
All three HOIST variants substantially reduce $|\Delta x|$ relative to VLA-50. Performance is not strictly monotonic in simulation: HOIST (+20 RL) achieves the lowest mean Manhattan error (2.54 cm), while HOIST (+30 RL) shows a slight increase (6.25 cm). We attribute this to evaluation variance over a small number of trials; on the real platform the mean Manhattan error decreases monotonically (Table [S2](#)).

Table S1. Terminal placement errors — Simulation. Values: mean $\pm$ std.

Method	$ \Delta x $ (cm)	$ \Delta y $ (cm)	$ \Delta \psi $ (°)	Manhattan (cm)
Human Expert Teleop	4.35 $\pm$ 2.67	3.00 $\pm$ 1.78	8.48 $\pm$ 3.99	7.35 $\pm$ 3.14
VLA-50	21.56 $\pm$ 1.20	0.88 $\pm$ 0.94	3.85 $\pm$ 1.94	22.44 $\pm$ 1.63
VLA-80	15.69 $\pm$ 2.18	2.89 $\pm$ 0.58	12.14 $\pm$ 1.69	18.58 $\pm$ 1.84
HOIST (+10 RL)	15.09 $\pm$ 3.53	1.83 $\pm$ 0.91	4.58 $\pm$ 1.97	16.92 $\pm$ 2.89
HOIST (+20 RL)	1.72 $\pm$ 0.57	0.82 $\pm$ 0.37	0.29 $\pm$ 0.27	2.54 $\pm$ 0.48
HOIST (+30 RL)	5.16 $\pm$ 7.11	1.09 $\pm$ 0.57	2.76 $\pm$ 2.04	6.25 $\pm$ 7.30



Manhattan error (Simulation).



Yaw error $|\Delta\psi|$ (Simulation).

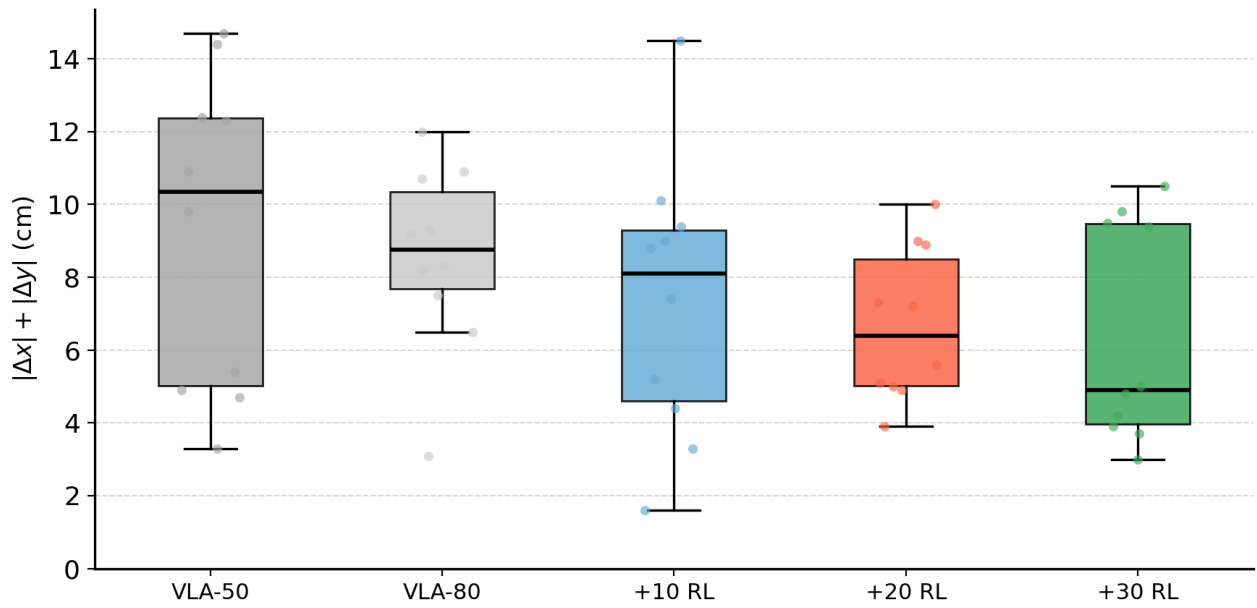
Figure S1. Simulation placement errors per trial (box = IQR, whiskers = $1.5 \times \text{IQR}$, dots = individual trials).

Real Platform

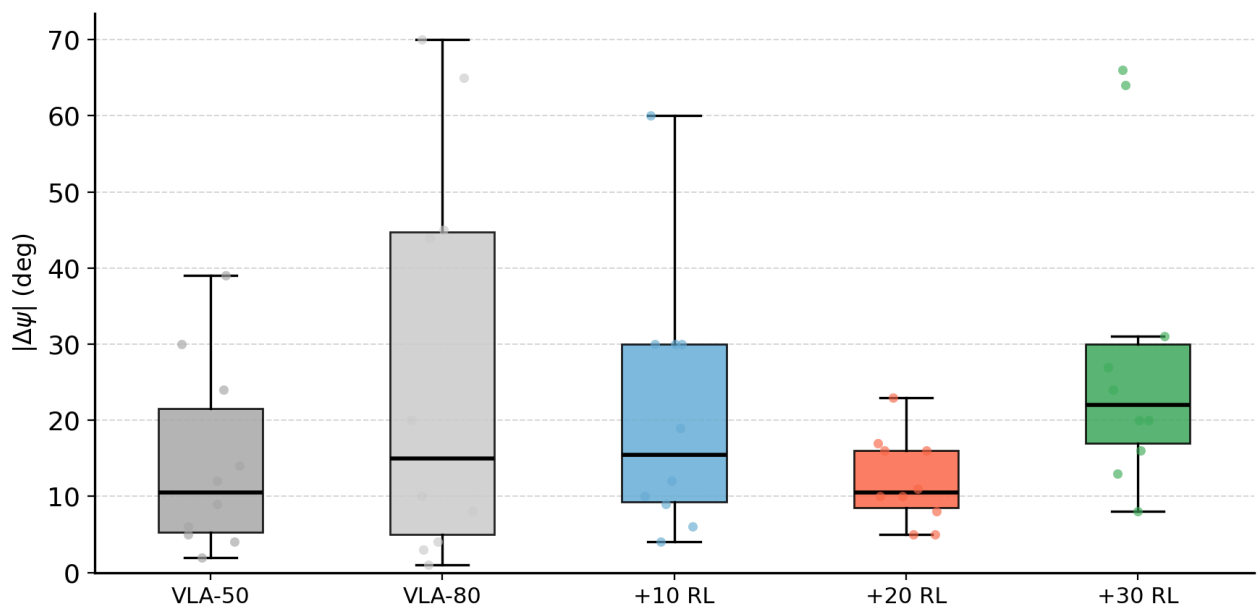
On the real platform, HOIST reduces mean Manhattan error from 9.28 cm (VLA-50) to 6.38 cm (+30 RL), indicating improved translational placement. However, yaw remains high-variance: +20 RL improves mean yaw relative to VLA-50, whereas +30 RL has larger mean yaw error. We therefore interpret the real-platform gains primarily as improvements in translational placement and Manhattan error, rather than as monotonic orientation refinement.

Table S2. Terminal placement errors — Real Platform. Values: mean ± std.

Method	$ \Delta x $ (cm)	$ \Delta y $ (cm)	$ \Delta \psi $ (°)	Manhattan (cm)
VLA-50	1.60 ± 1.46	7.68 ± 3.16	14.50 ± 12.46	9.28 ± 4.32
VLA-80	0.54 ± 0.29	8.03 ± 2.55	27.00 ± 26.64	8.57 ± 2.54
HOIST (+10 RL)	2.15 ± 1.50	5.22 ± 3.72	21.00 ± 17.09	7.37 ± 3.81
HOIST (+20 RL)	4.20 ± 1.26	2.49 ± 1.35	12.10 ± 5.78	6.69 ± 2.09
HOIST (+30 RL)	1.64 ± 0.98	4.74 ± 3.52	28.90 ± 20.15	6.38 ± 3.01



Manhattan error (Real Platform).



Yaw error $|\Delta \psi|$ (Real Platform).

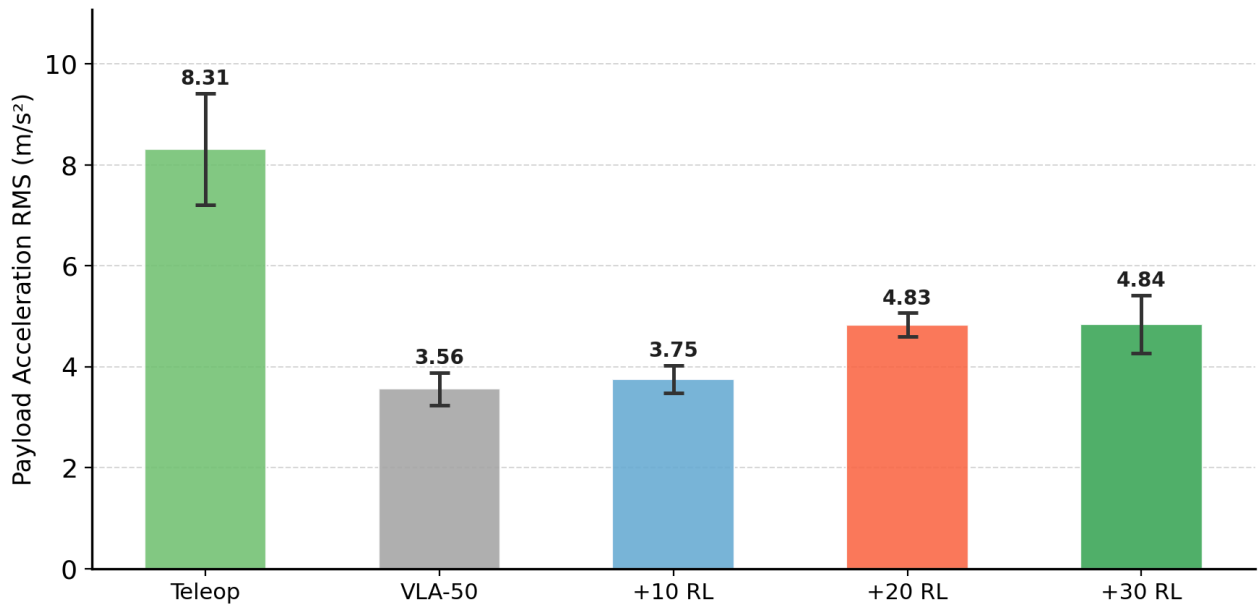
Figure S2. Real-platform placement errors per trial.

IMU Residual Motion Analysis

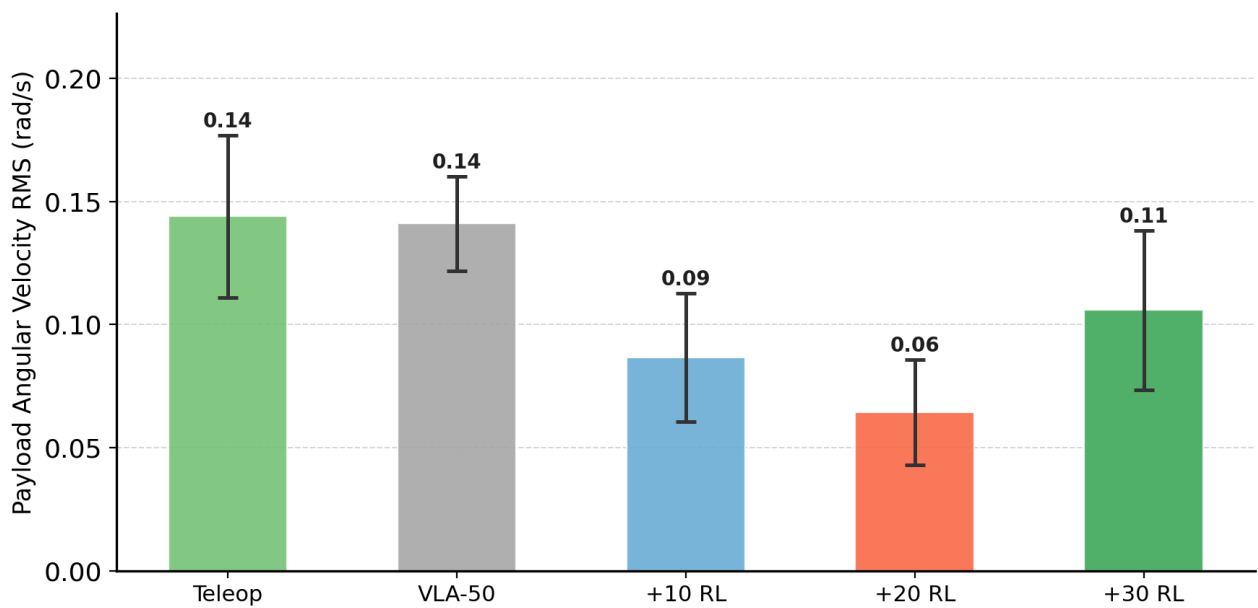
IMU data were collected in simulation only. We attach a 6-axis IMU to the payload and record accelerometer and gyroscope readings throughout each trial. Acceleration is gravity-compensated using the payload orientation quaternion and reported as RMS over active frames (frames where gravity-compensated acceleration magnitude $> 0.5 \text{ m/s}^2$ or angular velocity magnitude $> 0.01 \text{ rad/s}$).

Table S3. IMU residual motion — Simulation. Acceleration: gravity-compensated, active-frame RMS. Values: mean $\pm$ std.

Method	Acc (m/s ²)		Gyro (rad/s)	
	RMS	Peak	RMS	Peak
2-3 (lr)4-5				
Human Expert Teleop	8.31 $\pm$ 1.11	14.23 $\pm$ 1.20	0.144 $\pm$ 0.033	0.561 $\pm$ 0.116
VLA-50	3.56 $\pm$ 0.32	7.74 $\pm$ 0.00	0.141 $\pm$ 0.019	0.623 $\pm$ 0.050
HOIST (+10 RL)	3.75 $\pm$ 0.27	7.77 $\pm$ 0.11	0.087 $\pm$ 0.026	0.642 $\pm$ 0.135
HOIST (+20 RL)	4.83 $\pm$ 0.23	10.19 $\pm$ 3.96	0.064 $\pm$ 0.021	0.487 $\pm$ 0.184
HOIST (+30 RL)	4.84 $\pm$ 0.57	9.33 $\pm$ 1.75	0.106 $\pm$ 0.032	0.580 $\pm$ 0.106



Gravity-compensated acceleration RMS.



Payload angular velocity RMS.

Figure S3. IMU residual motion (Simulation). Error bars: ± 1 std.

Interpretation.

Compared with VLA-50, the HOIST variants generally reduce gyroscope RMS, with the strongest reduction at +20 RL. Acceleration metrics and gyroscope peaks are not uniformly improved across RL stages, so we interpret the IMU results as evidence for reduced angular RMS in some refined policies rather than a uniform reduction of all residual-motion metrics. Human teleoperation exhibits higher accelerometer readings, while gyroscope responses overlap with or are exceeded by some learned-policy settings.

Implementation Details and Hyperparameters

VLA Supervised Finetuning

Table S4. VLA supervised finetuning details.

Parameter	Value
Pretrained backbone	GR00T N1.6
VLM backbone	Eagle-Block2A-2B-v2
Finetuned modules	Final 4 VLM layers; flow-matching action expert; new-modality encoders
Frozen modules	Remaining VLM layers; whole-body controller
Input modalities	Ego RGB, ego depth, side-view RGB, proprioception, ego velocity, nav command, language goal
Action chunk horizon	24
Flow-matching denoise steps	4
Batch size	96
Learning rate	1×10^{-4}
LR schedule	Cosine
Warmup ratio	0.05
Warmup steps	500 steps
Weight decay	1×10^{-5}
Training steps	10,000
Equivalent epochs	~40

RL Rollout Refinement

Table S5. RL rollout refinement details.

Parameter	Value
Reward type	Terminal-only
Reward formula	$\exp(-[1.5(\Delta x/100\text{ mm})^2 + 1.0(\Delta y/80\text{ mm})^2 + 1.0(\Delta \psi /30^\circ)^2])$
RL rollouts per iteration	10
Total RL rollouts (HOIST)	30
Actor input	Observation o_t with cached VLM features
Actor output	Deterministic replacement initial-noise tensor
Noise tensor shape	24×25 (chunk horizon $\times$ command dimension)
Command dimensions	Head target; left/right end-effector targets; nav command; base height
Frozen modules	GR00T N1.6 backbone, SONIC WBC
Trainable modules	Noise-steering actor-critic
Batch size	256
Learning rate	3×10^{-4}
Gradient steps / iter.	20000

Hardware and Sensing

Table S6. Hardware and sensing setup.

Component	Specification
Robot platform	Unitree G1
Whole-body controller	SONIC (GR00T WBC, frozen)
VR teleoperation device	PICO VR headset and controllers
Payload size / mass	54 cm × 54 cm × 50 cm / 2.5 kg
Suspension length	35 cm
Target zone size	59 cm × 59 cm
Ego RGB/depth camera	Intel RealSense D435i
Side-view RGB camera	Intel RealSense D455
VLA control frequency	50 Hz
Trial reset procedure	Manual reset / warm-start from demonstration replay